

LOW-COST EEG USER GUIDE

Xavier Asghar - MEng Final Project - 2025/26

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1. OVERVIEW

This document contains the user guide for the Low-Cost EEG Module created as part of my University Final Year Project. The guide details how to setup and use the module for taking EEG measurements. The full details and design of the module can be found in the Final Report PDF.

In order to use the device, the procedure outlined below must be followed. Ensure the device is operated with battery power and the USB is disconnected at all times when connected to a subject. Each step is further explained in its respective section.

1. Ensure ESP32-S3 is flashed with the correct firmware. This includes specifying which channels will be in use to correctly derive the driven right leg bias signal.
2. Ensure USB is disconnected and the device is powered by a 3.7 V LiPo battery.
3. Prepare patient skin and apply electrodes.
4. Launch EEG_BLE_Receiver.exe (and press SW3 to take the board out of lead-off mode)
5. Launch Brainvision LSL Viewer to view EEG data in real time, or LabRecorder to record data to an XDF file.

Also contained in the repository are all the necessary production files (Gerbers, Schematic & Layout, BoM, STL files) to create your own module.



Figure 1: The complete EEG module

2. FLASHING THE MODULE

Before beginning EEG measurements the microcontroller must be running the correct code. This is especially important as the channels in use must be directly specified in the code to ensure the bias (driven right leg) signal is derived only from channels connected to the patient. The following steps outline how to correctly flash the module.

1. Install Visual Studio Code (VS Code).
2. Install the PlatformIO extension to VS Code.
3. In VS Code, open the Firmware_Platformio_Project folder, located in the software folder of the repository.
4. Wait for the project to finish configuring. This could take a few minutes.
5. Change the value of the BIAS_SENSP register, written to on line 102, in order to reflect the channels currently in use. The leftmost bit represents channel 8, while the rightmost bit represents channel 1. Channels in use should have a value of 1, while channels not in use should have a value of 0.

This ensures the driven right leg (bias) signal is only derived from channels connected to the subject. The example below shows the correct register value for when channels 1-3 are in use.

```
ADS.WREG(BIAS_SENSP, 0b00000111); // Derives bias from channels 1-3
```

6. Connect the module to the computer using the USB-C port, and upload the code, as shown in Figure 2. Ensure the correct COM port has been selected, and that the board power switch is set to USB power, as opposed to battery power.

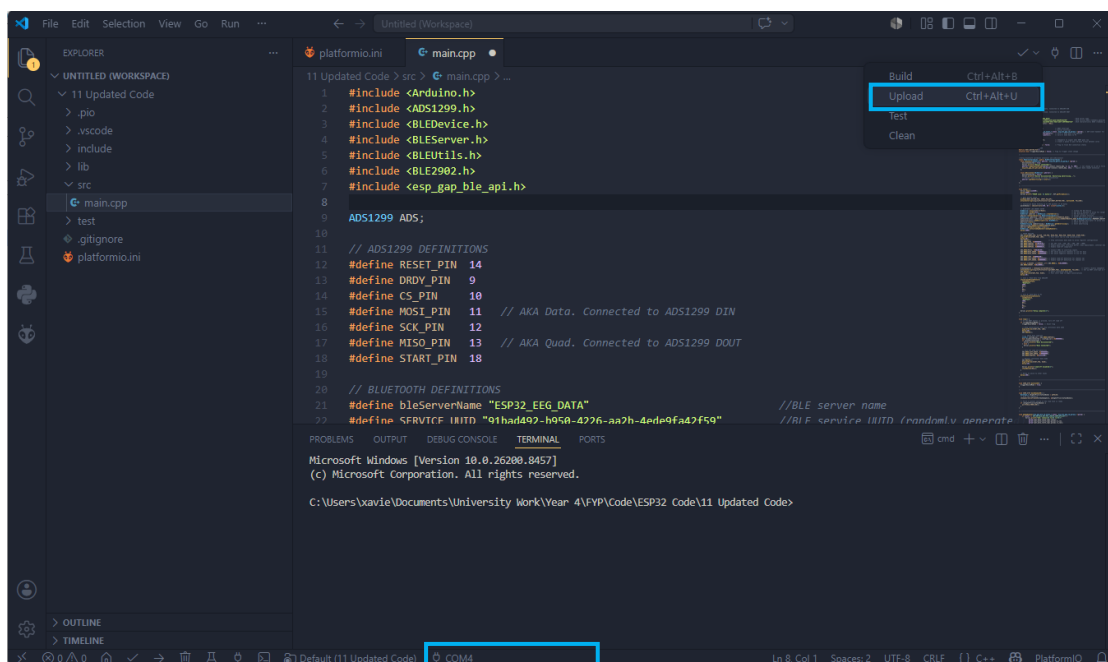


Figure 2: Platformio upload button

If the board is being flashed for the first time after assembly, the ESP32-S3 must be put into boot mode, using the following procedure:

- Press and hold the SW3 (BOOT) button
- Whilst continuing to hold SW3, press and release SW2 (RESET)
- Release SW3
- Upload code as shown in Figure 2.
- Press SW2

3. POWERING THE MODULE

When connected to a subject, the board should be powered from a 3.7 V LiPo battery, connected to the 2-pin JST PH connector highlighted in Figure 3. Ensure the polarity of the battery wires matches the polarity shown on the silkscreen.

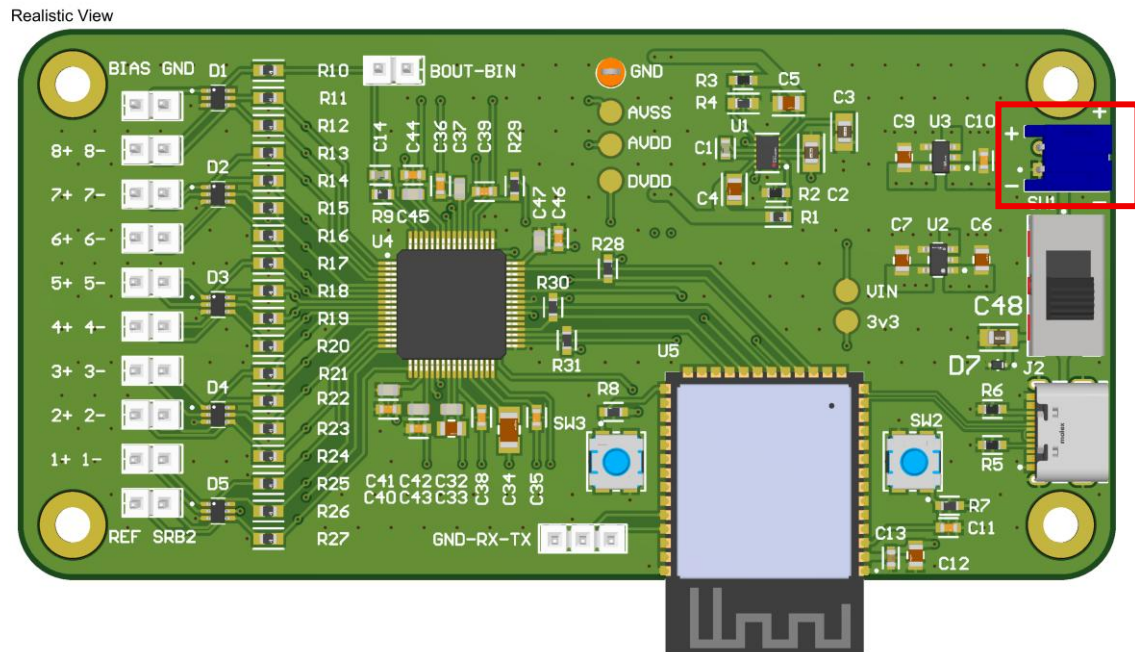


Figure 3: LiPo battery connector

During flashing, the board can be powered from a computer through the USB-C connector. Ensure that the USB-C cable is disconnected when the board is connected to a subject, to ensure electrical isolation of the subject from mains.

The switch (SW1) controls which power source is connected to the board, as well as acting as an on-off switch when the USB is not connected.

4. APPLYING ELECTRODES

The module has been designed to be used with standard gold-cup electrodes. As shown in Figure 4, the active electrodes should be attached to the **positive pin headers** (Pins 8+, 7+, ..., 1+), while the bias and reference electrodes should be attached to the **BIAS** and **REF** pins respectively.

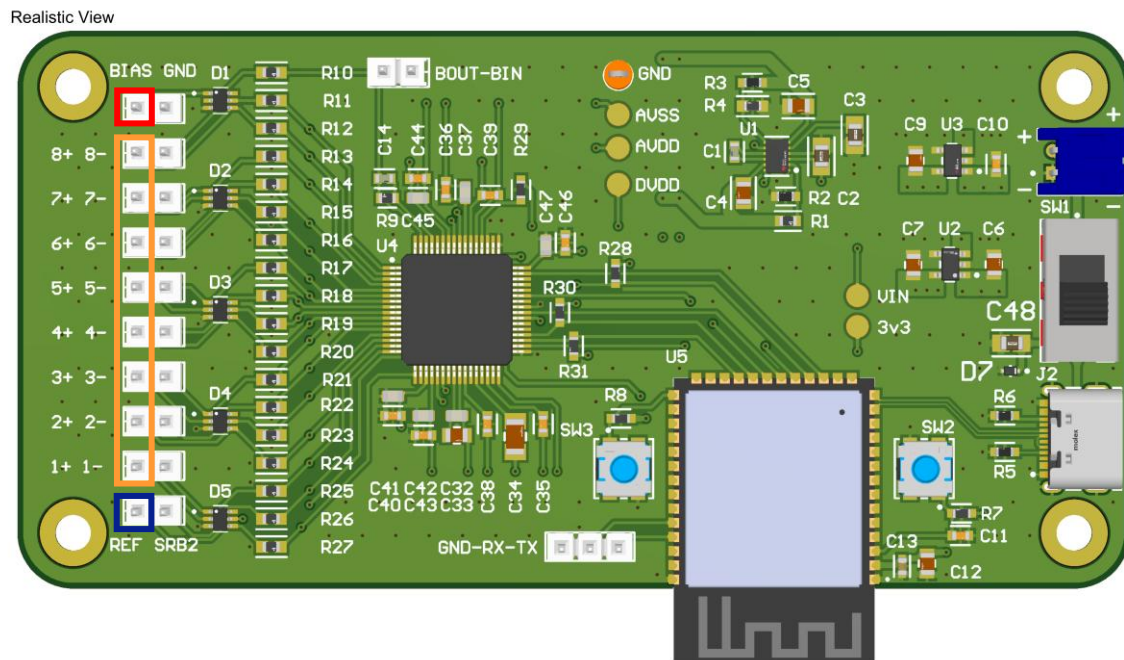


Figure 4: Electrode Connections

The active electrodes should be applied to the patient at locations set out by the international 10-20 system, while the reference and bias should typically be placed at electrically neutral locations, for instance the left and right mastoid, or the left and right earlobe. When applying electrodes to the patient, the following procedure should be adhered to:

1. Skin at electrode locations wiped with 70% Isopropyl Alcohol (IPA) wipes to remove oil and sweat.
2. The same locations rubbed with woven gauze swabs to remove dead skin cells, dirt, and residual IPA.
3. Electrode cups filled with Ten20 conductive paste to form a slight dome, such that when applied to the patient, some paste is pushed out from underneath the electrode.
4. Electrodes affixed to target locations. In hairless areas this can be done using medical tape. For areas containing hair, use tubular bandage, a headband, or a skull cap to maintain pressure on the electrodes.
5. Electrode wires taped to the patient (typically to the neck) with sufficient slack to prevent mechanical pulling and provide strain relief.

Alternatively, the IPA wipes and gauze can be replaced with NuPrep abrasive gel. The following guide can be used as reference for the application process:

<https://www.youtube.com/watch?v=b26vNbk8TOc>

If gold-cup electrodes with female Dupont style terminations cannot be sourced, it is possible to purchase electrodes with standard EEG touchproof connectors, which can then be cut-off, and replaced with Dupont style connectors, as shown in Figure 5. Here the resulting solderjoint has been covered with heatshrink.

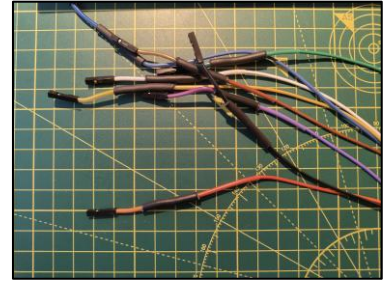


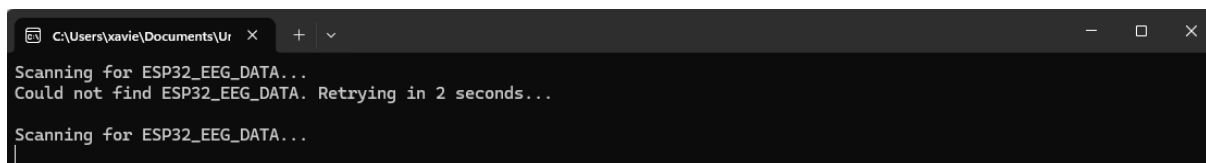
Figure 5: Electrode leads soldered to female Dupont wires

5. EEG BLE RECEIVER

In order to receive the EEG data being sent over bluetooth by the ESP32, and to subsequently push that data to the Lab Streaming Layer, the EEG_BLE_Receiver.EXE must be launched. Ensure that the computer/laptop running the application has Bluetooth enabled.

The application, shown in Figure 6, will scan for the ESP32, and once connected the following information is displayed on the console:

- **Dropped packets:** Counts the number of data packets that have been missed from the ADS1299. This counter should remain at 0 (however if the firmware is altered, and the ADS1299 is set to stream data at above 500 SPS, packets may be dropped).
- **Queue depth:** This should remain close to 0. If this number continually increases, it means that data is being read over bluetooth faster than it is being unpacked and pushed to the LSL.
- **Lead-Off bits:** These indicate the quality of the electrode connection for each channel. When in Lead-Off mode, a 0 indicates the electrode has a good connection, while a 1 indicates the electrode has a poor connection.
- **Channel voltages:** Displays the current voltage of each channel every 100 ms. This is to verify the system is working, rather than to be used as a way of reading EEG data.



```
C:\Users\xavie\Documents\Ur x + -
Scanning for ESP32_EEG_DATA...
Could not find ESP32_EEG_DATA. Retrying in 2 seconds...
Scanning for ESP32_EEG_DATA...
```

Figure 6: EEG_BLE_Receiver application

Upon reset, the EEG module starts in Lead-Off mode. This is where a small DC current is injected into each of the electrode inputs, and a voltage measured to determine if the electrode is on or off. This current can interfere with the EEG measurements, and thus **once the electrode connections have been verified, the SW3 button must be pressed to exit Lead-Off mode.**

6. VIEWING & RECORDING DATA

With the EEG_BLE_Receiver application running, the EEG data is now being pushed to the Lab Streaming Layer, where multiple free to use applications can be used to view and record the EEG data.

The most straightforward application to view the EEG data in real time is BrainVision LSL Viewer, which can be downloaded from the [Brain Products website](#). This application, shown in Figure 7, also lets you filter the EEG data, as well as view the FFTs for each channel. When viewing EEG data, I recommend using a mains frequency notch filter to remove power line interference, as well as a 0.5 – 150 Hz bandpass filter to remove noise.

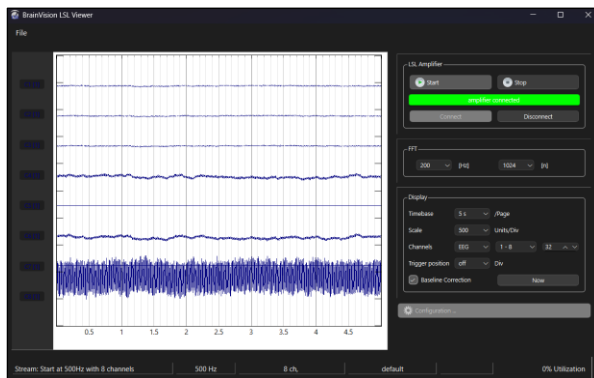


Figure 7.1: BrainVision LSL Viewer

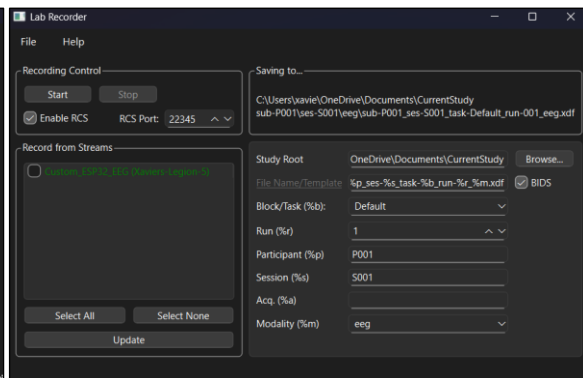


Figure 7.2: LabRecorder

Additionally, EEG data can be recorded using the [LabRecorder application](#), shown in Figure 7.2. This allows for the concurrent recording of multiple LSL streams, which is useful for linking external stimuli to their respective EEG responses, through the use of LSL marker streams. The EEG data is recorded into an XDF file, which can then be opened in python using the PyXDF library.